

Prebuilt

Printers that learn their errors from each other, without sharing a single measurement.

ScienceClaw · Manufacturing & Materials · MIT Media Lab

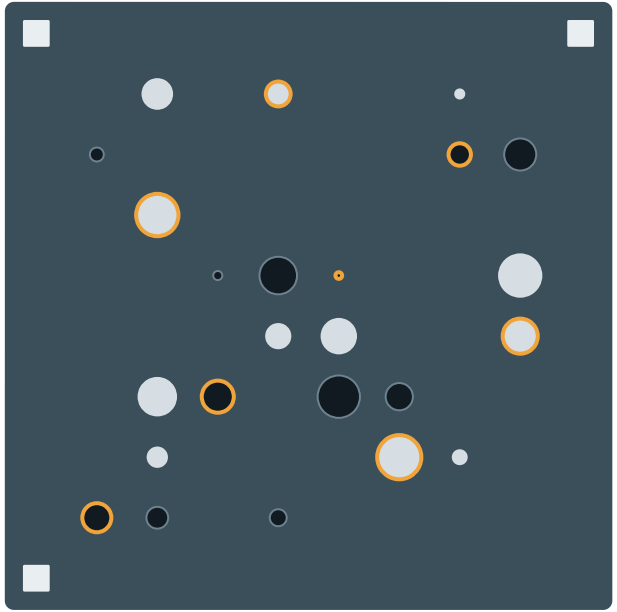
THE PROBLEM

Every shop re-learns its own printer

No printed part matches its file. Holes come out small, pins come out fat, by tens to hundreds of microns, and that decides whether parts fit.

The error belongs to your machine today: its resin, settings and wear. No datasheet knows it; you learn it by printing. Every site does that from scratch, and can't pool the data, because measurements expose process know-how and customer parts.

Can printers learn from each other without sharing a single measurement?



The test plate: pins and holes, 1–7 mm. Print it, measure it, and you know your printer.

WHAT WE BUILT

Two sites, one boundary, only beliefs cross

SLA

Resin site

prints, measures, learns its own model

⇄

μ, σ of 3 shared parameters

schema-checked · on a receipt

FDM

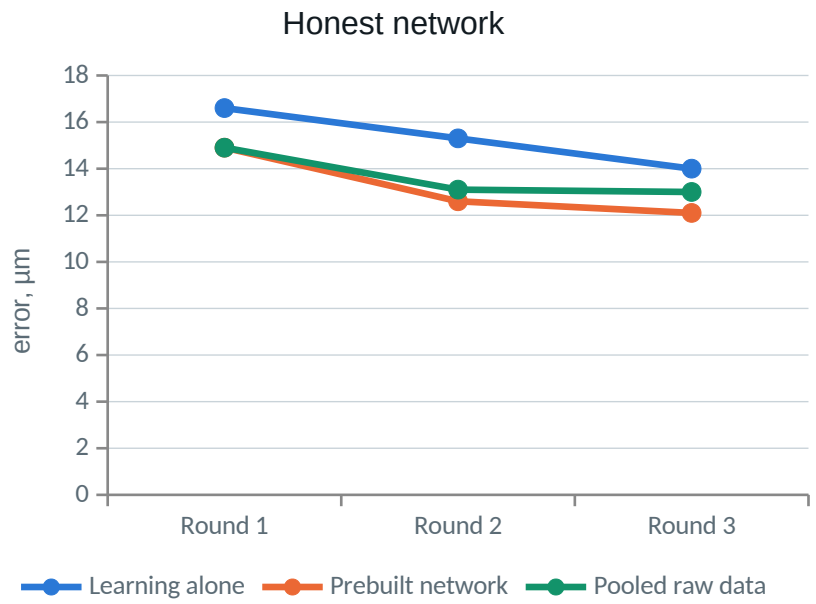
PLA site

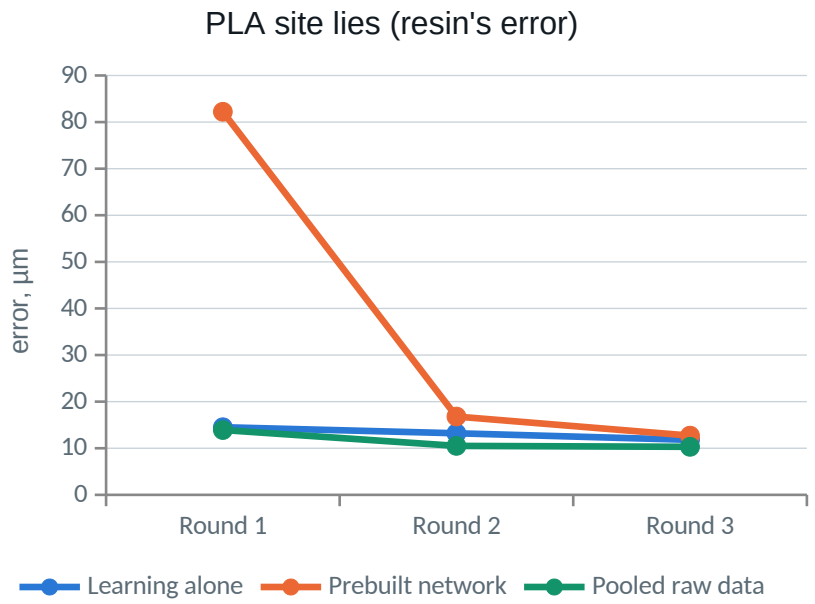
different machine, different physics

Stays home: every measurement, every machine-specific parameter, every customer part.

LIVE RUN

Watch the network learn and shrug off a liar





Held-out model error, µm, 20 simulated worlds, registered config. With a liar, resin's error spikes to 82.2 µm in round 1, trust collapses, and it is back to 12.7 µm by round 3. With the consistency check on, round 1 stays at 13.3 µm. Live view: viewer/demo.html.

RESULTS, IN SIMULATION

The win is many sites, not many agents

12%

lower error than learning alone, in 75 of 100 worlds. Pooling all the raw data: also 12%.

35%

with eight sites that each know little, better in 20 of 20 worlds. Many data-poor sites is where sharing pays.

0 rounds

for a liar: a consistency check catches its notes on arrival (180 of 180, 0 honest notes dropped).

Hidden-truth simulation, registered config, paired per world. The consistency check is an added option, off in the registered config. The agent layer adds nothing measurable. At two sites the gain is 1.6 µm, below what a real part can show, so we claim it in simulation only.

RIGOR

A real printer for scale, simulation to score

Real printer

Formlabs Form Studio 1: a 1.5 mm pin measured 1.41–1.48 mm by micrometer, 24-89 µm small.

Vendor spec

±0.08 mm on 1–30 mm features. Real error is tens of µm, as the spec says.

Simulator

Simulated resin pins miss by −0.1 to +0.27 mm. The real pin is inside, at the small edge: each printer must learn its own error.

Why not real

The sharing gain is 1.6 µm. Two readings of one pin differed by 65 µm. Only hidden truth can score it.

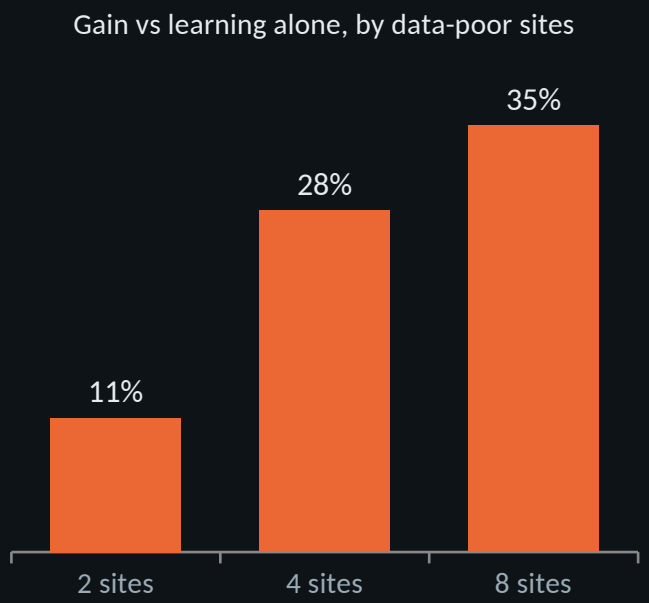
Registered before printing: two printers can't show the sharing gain on real parts. Negative control: when printers share no physics, sharing gains nothing, so the gain isn't built into the scoring.

WHO IT'S FOR

Fleets where fit decides and data can't leave

Dental labs	Patient scans. Crowns, models and guides must seat on the first print.
Print farms, service bureaus	Customer geometry is the customer's IP. Many machines, every job must fit.
Distributed manufacturing	One part, many vendors. Each vendor's process is its edge.
Next: microfluidics, bioprinting	Smaller features, costlier tests, stricter data rules.

What they get: parts that fit the first time (corrected designs: size error 132 → 9 µm), and sharing that stays safe when printers differ.



Print anywhere, trust the part.

github.com/matte-realize/prebuilt · all numbers simulation

Many sites beat one; agents don't beat random

Round 3, paired simulated worlds	Gain	Worlds better	p
Network vs single site alone	12.3% (1.6 µm)	75 / 100	<0.001
Pooled raw data (upper bound) vs alone	12.2%	64 / 100	
Agents only, messages cut, vs alone	2.4%	56 / 100	0.27
Agent auction vs random pieces	+1.0%	31 / 60	0.90
4 / 8 data-poor sites vs alone	28.5% / 34.7%	35 / 40 · 20 / 20	<0.001
Negative control: nothing shared	−13.0% (adaptive spread: −0.4%)	19 / 60	0.006

Hidden-truth simulator, registered config, error against true size on held-out pieces, sign test over worlds. 90% intervals cover 90–91%. GPU posterior up to 84× NumPy on an H100. Reproduce every row: docs/validation-and-benchmarks.md.